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Domain 1 - Solid Mechanics

Element type used: two node, two DOF per node, one axial DOF

Element stiffness = $k = \frac{AE}{L}$, $K = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

Problem Statement:

A bi material thermal actuator strut sits inside a Satellite deployment latch. The strut is clamped to a rigid bulkhead at one end. Its free tip forces a compliant compliant end stop (a leg spring pad) that is meant to strike once only when the assembly is hot enough. A preload rod pulls back on the mid joint of the strut opposing the expansion, so contact is not guaranteed.

Geometry and material data:

Element	Material	L (mm)	A (mm ²)	E (MPa)	α (1/c)	ΔT (°C)
1	Ti-6AL-4V (Titanium)	200	400 (20x20)	110000	9×10^{-6}	140°
2	AL6061-T6 (Aluminium)	100	200 (20x10)	7000	23×10^{-6}	80°

leg spring stop - $K_s = 50000 \text{ N/mm}$

The temperature rise is non uniform because the aluminium segment sits next to the radiator patch, treat each element as being its own uniform ΔT

Boundary condition:

node 1: $u_1 = 0$

node 4: $u_4 = 0$

node 3 to node 4: Initial clearance $\delta = 0.25 \text{ mm}$

E1 carries load only when $u_3 \geq \delta$

Stop is unilateral (compression only) and frictionless

Loading:

Thermal $\Delta T_1 = 140^\circ\text{C}$, $\Delta T_2 = +80^\circ\text{C}$, stress free at reference state.

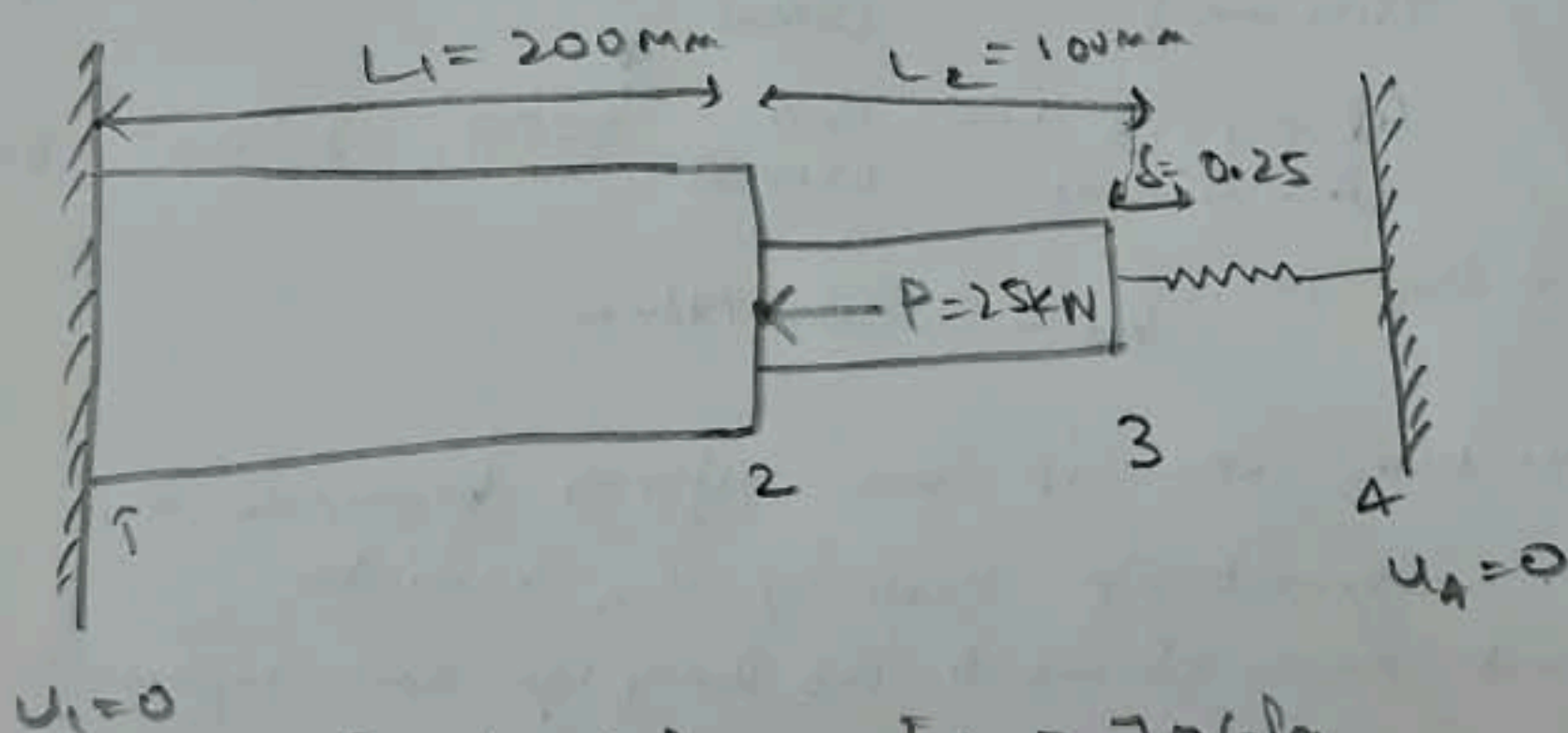
Mechanical $P = 25 \text{ kN}$, applied at node 2

directed toward wall (-x)

Self weight, buckling and creep neglected

Required:

- ① Determine whether the tip contacts the stop
- ② Nodal displacement u_2, u_3
- ③ Axial force and stress in each bar element.
- ④ Force in stop and reaction in bulk head and verify global equilibrium



$$E_1 = 110 \text{ GPa}$$

$$A_1 = 400 \text{ mm}^2$$

$$\alpha_1 = 9 \times 10^{-6} / ^\circ\text{C}$$

$$E_2 = 70 \text{ GPa}$$

$$A_2 = 200 \text{ mm}^2$$

$$\alpha_2 = 23 \times 10^{-6} / ^\circ\text{C}$$